

- Each customer order documented in MongoDB collection
 - order ID
 - customer information
 - order items
 - order metadata
 - order status
- Nested document used to store information about the delivery route.
- Incorporating Geographic coordinates (latitude and longitude) using MongoDB's geospatial indexing.



Aerial Delivery Solutions

MongoDB

Spatial Considerations

- optimized route planning and tracking the movement of deliveries through various waypoints.

Aerial Delivery to enhance AGM delivery processes

Benefits and Limitations

- Benefits**
- Handle diverse and dynamic data associated with aerial deliveries. The ubiquitous nature of MongoDB accommodates changes in the order structure.
 - Different tracking sensors equip drones to specific orders or routes, and geospatial capabilities facilitate spatial queries for location-based analysis.

- Limitations**
- Real-Time Precision:** Current tracking systems may struggle to provide real-time and precise location data for aerial deliveries.
 - Spatial Complexity:** Aerial deliveries introduce complexities in route planning and spatial coordination. This can get more complex considering FAA requirements/permissions, weather conditions, and potential concerns when personal information, order details, and geospatial location data are collected.

MongoDB

Data Flexibility

- The data associated with aerial deliveries is diverse and can evolve over time.

MongoDB

Once Upon A Time On the Bart

Jacob Brinkerhoff, Kevin Kuc, & Maria Lee



Neo4j

BART-based Delivery System

- enhance AGM delivery system utilizing BART for pick up or deliveries
- optimize routes or staging locations
- order process can adapt quickly to changes in routes and schedules
- offers advanced graph management capabilities to AGM data science team



Louvain Modularity

- for adding BART stations as pickup locations
- grouping stations to create main pick up location per group
 - not feasible to have all BART stations be pick up spots

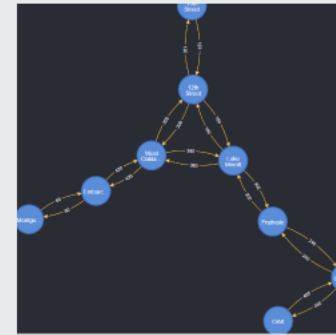
Closeness Centrality

- for using BART to transport deliveries
- determining most centralized station for a single central staging location

Dijkstra

- for using BART to transport deliveries
- generating accurate delivery time estimates

graph algorithms



Questions?

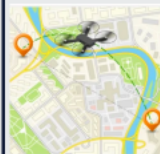
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Utilizing Redis for Real Time Tracking

- key-value store : optimized for high speed operation
- live inventory management
- order status from initial order, packing, to delivery
- store and update geographical locations of delivery drones
- drone maintenance

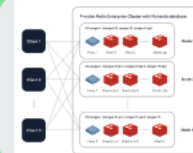
Redis



Redis

Advantages of Redis

- Ultra fast data processing
- reduced latency in information retrieval
- flexibility with unstructured data and scalability
- concurrent data handling



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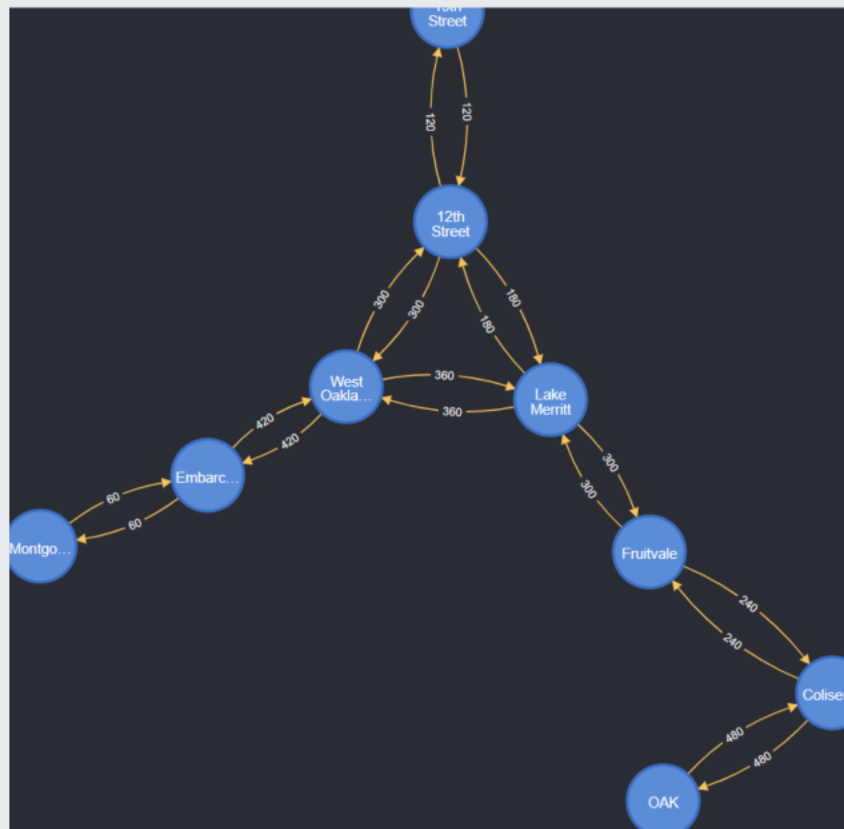


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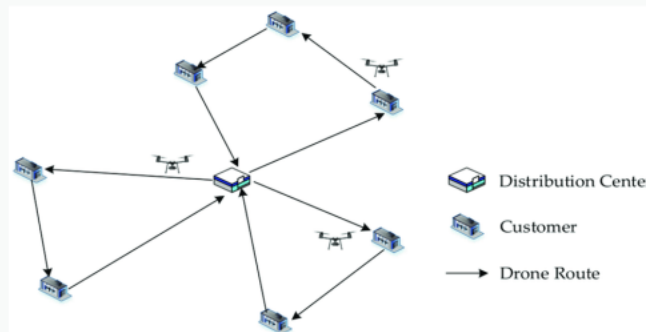
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Aerial Delivery Solutions

MongoDB

Benefits and Limitations

Benefits

- Handle diverse and dynamic data associated with aerial deliveries.
- The schema-less nature of MongoDB accommodates changes in the order structure.
- Efficient indexing ensures quick access to specific orders or routes, and geospatial capabilities facilitate spatial queries for location-based analysis.

Limitations

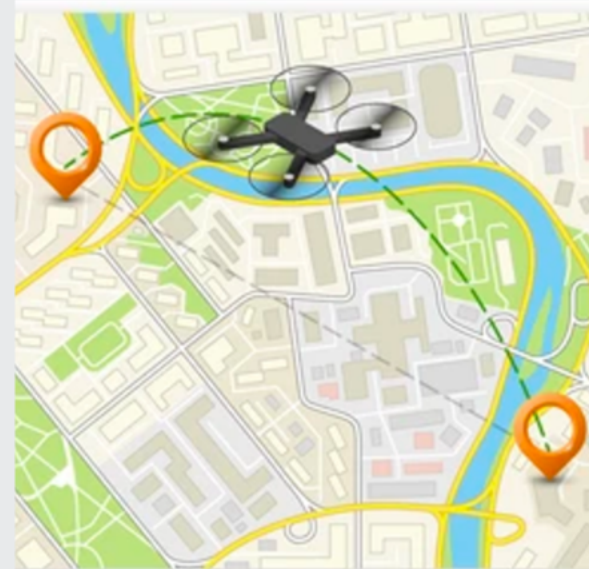
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MongoDB

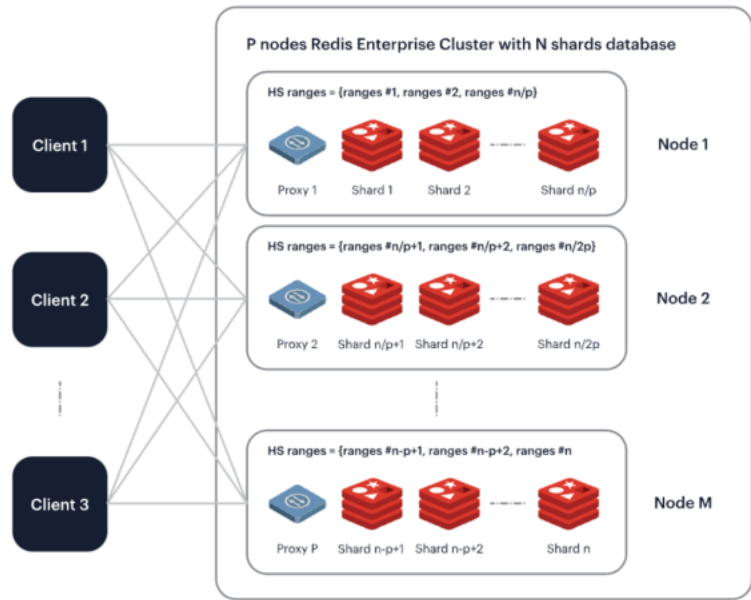
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